

LECTURE 19 — N-VARIATE JOINT DISTRIBUTIONS

High-Density Logical Revision Map

1 WHY JOINT DISTRIBUTIONS? (Concept Foundation)

Step 1: Representation

$$\mathbf{X} = [X_1, X_2, \dots, X_N]^T$$

Reasoning:

- Instead of treating variables separately, we group them.
- Joint distribution describes probability behavior of all variables together.
- Transition: single-variable $\rightarrow$ multi-variable framework.

Step 2: Forms of Representation

- Joint PMF (discrete)
- Joint PDF (continuous)
- Joint CDF

Reasoning:

- Direct extensions of single-variable definitions.
- Now we consider simultaneous events.

2 CORE DEFINITIONS

2.1 Joint PMF

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

Reasoning:

- Probability of all variables taking values simultaneously.
- Represents one combined event.

2.2 Joint CDF

$$F(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

Reasoning:

- Accumulates probability over an N-dimensional region.
- Extension of 1D CDF: line $\rightarrow$ region.

2.3 Joint PDF

$$f = \frac{\partial^N F}{\partial x_1 \cdots \partial x_N}$$

Reasoning:

- Same logic as 1D: PDF = derivative of CDF.
- For N variables, take N partial derivatives.

3 INDEPENDENCE CONDITION

$$f(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

Reasoning:

- Joint PDF equals product of marginals.
- This is the test for independence.

4 EXAMPLE 1 — JOINT GAUSSIAN

Step 1: Given

$$\begin{aligned}\mu_X &= 2, & \sigma_X^2 &= 1 \\ \mu_Y &= -3, & \sigma_Y^2 &= 4 \\ \text{cov}(X, Y) &= -1\end{aligned}$$

Step 2: Correlation

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Reasoning: Gaussian formula uses correlation.

Step 3: Standard Deviations

$$\sigma_X = 1, \quad \sigma_Y = 2$$

Step 4: Compute ρ

$$\rho = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

Step 5: Joint PDF

Substitute values into Gaussian formula.

Step 6: Marginals

$$f_X(x) = \int f_{XY}(x, y) dy, \quad f_Y(y) = \int f_{XY}(x, y) dx$$

$$X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2)$$

Reasoning: Marginals of jointly Gaussian variables are Gaussian.

Step 7: Compute $P(X < 0)$

$$P(X < 0) = F_X(0)$$

Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$P(X < 0) = P\left(Z < \frac{0 - 2}{1}\right) = P(Z < -2)$$

Symmetry

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

$$P(Z < -2) = P(Z > 2)$$

Q-function

$$Q(x) = P(Z > x)$$

$$P(X < 0) = Q(2)$$

5 APPLICATION — LARGE MIMO SYSTEM

System

- M transmit antennas
- N receive antennas

Channel Model

$$h_{ij} = \text{random variable}$$

Matrix Form

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$

Reasoning: Multiple random variables $\Rightarrow$ joint modeling.

6 EXAMPLE 2 — UNIFORM 3D VECTOR

Given

$$|x_1| \leq 1, \quad |x_2| \leq 1, \quad |x_3| \leq 1$$

$$f = \begin{cases} C, & \text{inside cube} \\ 0, & \text{otherwise} \end{cases}$$

Find C

$$\iiint f = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

Marginals

$$f_{X_1, X_2} = \int_{-1}^1 \frac{1}{8} dx_3 = \frac{1}{4}$$

$$f_{X_1} = \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3 = \frac{1}{2}$$

Conditional PDFs

$$f_{X_1|X_2,X_3} = \frac{1/8}{1/4} = \frac{1}{2}$$

$$f_{X_1,X_2|X_3} = \frac{1/8}{1/2} = \frac{1}{4}$$

Independence Check

$$\text{LHS} = \frac{1}{8}$$

$$\text{RHS} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion: Independent

7 EXAMPLE 3 — TRIVARIATE EXPONENTIAL

Given

$$f_{XYZ} = Ke^{-(ax+by+cz)}, \quad x, y, z > 0$$

Find K

$$\iiint f = 1$$

$$e^{-(ax+by+cz)} = e^{-ax}e^{-by}e^{-cz}$$

$$K \left(\frac{1}{a}\right) \left(\frac{1}{b}\right) \left(\frac{1}{c}\right) = 1$$

$$K = abc$$

Marginals

$$f_{XY} = abe^{-(ax+by)}$$

$$f_X = ae^{-ax}$$

Independence Check

$$\text{LHS} = abce^{-(ax+by+cz)}$$

$$\text{RHS} = (ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion: Independent

8 FINAL REVISION FLOW (MENTAL MAP)

- **Vector Form** $\rightarrow$ Joint Behavior
- **PMF / CDF / PDF** $\rightarrow$ Core Definitions
- **Independence** $\rightarrow$ Product Rule
- **Gaussian Case:**
 - cov $\rightarrow \rho$
 - Substitute into joint PDF
 - Marginals
 - Standardization
 - Q -function
- **Uniform Cube:**
 - Normalize
 - Integrate (marginals)
 - Conditional PDFs
 - Independence check
- **Exponential Case:**
 - Factorization
 - Normalize
 - Marginals
 - Independence check